LAB-IV TASK 1 lab Report

Student Information:

Hikmat Ullah Jan

Electrical Eng (cammunicaton)

Registration No: BF25NWELE0662

Semester 2nd

Section B

Title:

Grade Evaluation Using Switch Statement

Objective:

To create a program that takes a character grade and displays its description.

Theory:

Switch statements can also work with character values. Each case represents a specific grade.

Program (with comments):

```
#include <iostream>   // Input-output library
using namespace std;  // Standard namespace

int main()
{
    char grade; // Variable to store grade

    cout << "Enter your grade (A, B, C, D, F): "; // Prompt user
    cin >> grade; // Take input

    switch(grade) // Switch on character
    {
        case 'A':
            cout << "Excellent" << endl;
            break;

        case 'B':
            cout << "Very Good" << endl;
            break;

        case 'C':
            cout << "Good" << endl;
            break;

        case 'D':
            cout << "Pass" << endl;
            break;

        case 'F':
            cout << "Fail" << endl;
            break;

        default:
            cout << "Invalid Grade" << endl; // Invalid input
    }

    return 0; // End program
}
```

Detailed Explanation:

The program accepts a character input representing a grade. The switch statement checks the grade and prints the corresponding message.

If the user enters a value outside the given options, the `default` case handles it.

Output (Example):

Enter your grade: B
Very Good

Conclusion:

Switch statements can efficiently handle character-based conditions and improve code clarity.